

Messy (wide format)

Department	January	February	March
Sales	\$33,136.00	\$33,136.00	\$33,136.00
Administration	\$27,319.48	\$27,319.48	\$27,319.48
Marketing	\$20,646.36	\$20,646.36	\$20,646.36



Reshape to tidy format

Tidy (long format)

Department	Month	Budget Amount
Sales	January	\$33,136.00
Sales	February	\$33,136.00
Sales	March	\$33,136.00
Administration	January	\$27,319.48
... 9 rows in all: 3 departments × 3 months		

Amounts: Charles River Budget table, salaries expense budgets for fiscal year 2025.

Tidy data principles violated:

1. Month is stored in column headers, not in its own column.
 2. Each row contains three observations (three months), not one.
- This layout prevents direct use in PivotTables, SQL GROUP BY queries, or Power BI data models.

Tidy data principles satisfied:

1. Each variable (Department, Month, Budget Amount) has its own column.
2. Each observation (one department in one month) has its own row.
3. This single table holds one type of observational unit (monthly budgets).